# Answer Key & Teacher Notes

Session 1 — Addition Equations · Session 2 — Subtraction Equations

## HOW TO USE THIS LESSON'S PAGES

*one lesson = two sessions*

| | |
|---|---|
| **Warm-Up** | About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key. |
| **Prerequisite Practice A/B** | Before the lesson, for students who need the skill the session uses. Key in that folder. |
| **Guided Notes** | During the lesson. Students fill in the blanks as you teach; answers below. |
| **Practice Form A** | In class, after the lesson. Two pages. The self-check code tells students if they are on track. |
| **Practice Form B** | Same skills, new numbers: reteach, retake, small group or absent students. |
| **Exit Ticket** | Last 5 minutes. Two per page; cut in half. |
| **Homework Packet** | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 · TEACHER NOTES

*Addition Equations*

### TALK PROMPTS

- Why do you subtract instead of add when the variable has a number added to it?
- How does the story tell you which number is added and which number is the total?

### PARTNER TASK

Form A, problems 4 and 5. One partner solves Aquarium Fish (f + 14 = 62); the other checks by substituting. Switch roles with Savings Jar (d + 16 = 73).

### WATCH FOR

- Subtracting from one side only (b + 26 = 40 gives b = 40).
- Adding the number instead of subtracting it (y + 34 = 50 gives y = 84).
- Misaligning decimal points (5.9 − 2.4).

## SESSION 2 · TEACHER NOTES

*Subtraction Equations*

### TALK PROMPTS

- Why does adding the same amount to both sides not change the solution?
- Why does adding back the amount used bring you to the starting amount?

### PARTNER TASK

Form A, problems 3 and 4. One partner solves Phone Storage (g − 12 = 29); the other checks by substituting. Switch roles with Library Shelf (b − 38 = 54).

### WATCH FOR

- Subtracting again instead of adding (k − 19 = 33 gives 14).
- Accepting an answer smaller than the number on the right side of x − a = b.
- Misaligning decimal points with money ($18.50 + $31.75).

## ITEM ANALYSIS

*suggested DOK; judge the reasoning required, not the question label*

| Sheet | Standard | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **S1 · Form A** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |
| **S1 · Form B** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |
| **S2 · Form A** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 2 |
| **S2 · Form B** | 6.EE.B.7 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 1 | 1 |

*Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.*

### GUIDED NOTES

**inverse operations:** Two operations that **undo** each other.

**isolate:** Get the variable **alone** on one side.

**Subtraction Property of Equality:** Subtract the same number from **both** sides. They stay equal.

- Subtraction **undoes** addition.
- To solve $x + 14 = 40$, subtract **14** from both sides so the equation stays **balanced**.

**Example:** Subtract **21** from both sides. $p = 50 - 21 = \mathbf{29}$. Check: $21 + 29 = \mathbf{50}$. Answer: $p = \mathbf{29}$

### YOU TRY

**1.** a. x = 16 b. m = 18

**2.** k = 10 − 4.5 = 5.5. Check: 5.5 + 4.5 = 10.

### EXIT TICKET

**1.** a. a = 27 b. k = 4.5

**2.** b + 25 = 71; b = 46 books

**3.** **B** (subtract 18 from both sides)

### PRACTICE FORM A

**1.** a. n = 37 b. p = 25 c. x = 28 d. y = 12.5 e. m = 18 f. w = 6.5

**2.** a. r + 19 = 56 b. r = 56 − 19 = 37 c. 37 + 19 = 56

**3.** s + 3.5 = 12.5; s = 9 miles

**4.** f = 62 − 14 = 48 fish

**5.** d = 73 − 16 = 57 dollars

**6.** r + 2.4 = 5.9; r = 3.5 inches

**7.** p + 38 = 91; p = 53 pages. Check: 53 + 38 = 91.

**8.** a. s + 46 = 128 b. s = 82 dollars c. 128 + e = 200; e = 72 dollars

**9.** He must subtract 26 from both sides to keep the equation balanced: b = 40 − 26 = 14. Check: 14 + 26 = 40.

**10.** **B** ($y = 27$)

**11.** **C** (subtract the same number from both sides)

**Self-check word** **BALANCE**

### PRACTICE FORM B

**1.** a. n = 37 b. p = 27 c. x = 33 d. y = 11.5 e. m = 26 f. w = 7.5

**2.** a. r + 23 = 61 b. r = 61 − 23 = 38 c. 38 + 23 = 61

**3.** s + 4.5 = 14.5; s = 10 miles

**4.** f = 63 − 17 = 46 fish

**5.** d = 81 − 18 = 63 dollars

**6.** r + 1.8 = 6.3; r = 4.5 inches

**7.** p + 42 = 97; p = 55 pages. Check: 55 + 42 = 97.

**8.** a. s + 54 = 135 b. s = 81 dollars c. 135 + e = 200; e = 65 dollars

**9.** Adding 34 does not undo + 34. He should subtract 34 from both sides: y = 50 − 34 = 16. Check: 16 + 34 = 50.

**10.** **B** ($b = 15$)

**11.** **C** (balanced)

**Self-check word** **INVERSE**

## SESSION 2 · ANSWERS

### GUIDED NOTES

**Addition Property of Equality:** Add the same number to **both** sides. They stay equal.

**difference:** The answer when you **subtract**.

**isolate:** Get the variable **alone** on one side.

- Addition **undoes** subtraction.
- To solve $p - 7 = 12$, add **7** to both sides.

**Example:** Add **9** to both sides. $m = 14 + 9 = \mathbf{23}$. Check: $23 - 9 = \mathbf{14}$. Answer: $m = \mathbf{23}$

### YOU TRY

**1.** a. x = 26 b. n = 55

**2.** p = 4 + 2.5 = 6.5. Check: 6.5 − 2.5 = 4.

### EXIT TICKET

**1.** a. n = 42 b. p = 9.5

**2.** s − 87 = 26; s = 113 snacks

**3.** **B** (add 19 to both sides)

### PRACTICE FORM A

**1.** a. n = 47 b. p = 14 c. k = 65 d. m = 45 e. r = 10 f. y = 75

**2.** a. p − 47 = 16 b. p = 16 + 47 = 63 c. 63 − 47 = 16

**3.** g − 12 = 29; g = 41 gigabytes

**4.** b = 54 + 38 = 92 books

**5.** n = 89 + 145 = 234 miles

**6.** f − 18.50 = 31.75; f = $50.25

**7.** a − 128 = 95; a = 223 pounds. Check: 223 − 128 = 95.

**8.** a. b − 34 = 89 b. b = 123 books c. 89 − 45 = r; r = 44 books

**9.** Subtracting 19 does not undo − 19. She should add 19 to both sides: k = 33 + 19 = 52. Check: 52 − 19 = 33.

**10.** **B** ($m = 55$)

**11.** **C** (greater than 30)

**Self-check word** **ISOLATE**

### PRACTICE FORM B

**1.** a. n = 43 b. p = 11 c. k = 81 d. m = 55 e. r = 7 f. y = 85

**2.** a. p − 38 = 24 b. p = 24 + 38 = 62 c. 62 − 38 = 24

**3.** g − 15 = 34; g = 49 gigabytes

**4.** b = 57 + 42 = 99 books

**5.** n = 75 + 160 = 235 miles

**6.** f − 24.25 = 30.50; f = $54.75

**7.** a − 136 = 89; a = 225 pounds. Check: 225 − 136 = 89.

**8.** a. b − 29 = 96 b. b = 125 books c. 96 − 40 = r; r = 56 books

**9.** Subtracting 24 does not undo − 24. He should add 24 to both sides: x = 60 + 24 = 84. Check: 84 − 24 = 60.

**10.** **B** ($x = 26$)

**11.** **A** (add the same number to both sides)

**Self-check word** **SOLVING**

## HOMEWORK PACKET · ANSWERS

*same answers for the core packet and Level 1*

### SHEET 1 · ADDITION EQUATIONS

**1.** a. x = 8 b. n = 18 c. y = 35 d. m = 37 e. k = 5.5 f. w = 4.25

**2.** p + 46 = 93; p = 47

**3.** b + 27 = 64; b = 37 books. Check: 37 + 27 = 64.

**4.** 38.50 + m = 52; m = $13.50. Check: 38.50 + 13.50 = 52.

**5.** No. n = 32 is correct. Dante checked with the wrong operation. He should substitute 32 into the original equation: 32 + 18 = 50, which is true.

### SHEET 2 · SUBTRACTION EQUATIONS

**1.** a. n = 45 b. k = 85 c. p = 9.5

**2.** t − 54 = 35 (or t − 35 = 54); t = 89

**3.** g − 18.25 = 31.75; g = $50.00

**4.** c − 26 = 59; c = 85 cars. Check: 85 − 26 = 59.

**5.** Leo. 84 − 24 = 60 is true. Mia subtracted instead of adding: 36 − 24 = 12, not 60.

**6.** **C** ($p = 74$)

**7.** **B** ($m - 48 = 37$)

# October 2026 conceptual additions — teacher key

## Session 1 · Addition Equations

Subtract 8 from both sides: x=15. Check 15+8=23. Equal removal preserves balance; a one-sided change need not.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: Subtract 8 from both sides: x=15. Check 15+8=23. Equal removal preserves balance; a one-sided change need not.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

## Session 2 · Subtraction Equations

First add 7 to both sides: n=25. Second subtract 7: n=11. The inverse operation depends on the original operation.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: First add 7 to both sides: n=25. Second subtract 7: n=11. The inverse operation depends on the original operation.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.